

CONTEMPORARY REVIEW

Transcatheter Aortic Valve Durability: Focus on Structural Valve Deterioration

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ABSTRACT: Transcatheter aortic valve replacement has emerged as a valuable alternative to surgical aortic valve replacement in patients with severe aortic stenosis. Given the expansion of transcatheter aortic valve replacement to lower-risk and younger populations with longer life expectancy, the durability of transcatheter heart valves (THVs) has become an important issue that may impact cardiovascular outcomes. THVs share similarities with surgical valves but have unique features, including a trend to larger effective orifice area and less prosthesis–patient mismatch, interactions with the native valve, and crimping process, that may all potentially influence a THV's life span. Multiple mechanisms may lead to bioprosthetic valve dysfunction, including structural valve deterioration, thrombosis, endocarditis, and nonstructural valve deterioration. With an incidence of up to 12.3% 5 years after transcatheter aortic valve replacement, structural valve deterioration represents the ultimate consequence of fibrotic remodeling and calcification within the bioprosthesis, driven by thrombotic and inflammatory processes involving the native aortic valve and influenced by patient and procedural factors. Understanding these mechanisms is crucial for improving THV durability.

Key Words: bioprosthetic valve failure ■ durability ■ structural valve deterioration ■ transcatheter aortic valve replacement ■ transcatheter heart valve

Two decades after the “first-in-man case,” transcatheter aortic valve replacement (TAVR) has become a robust alternative to surgical aortic valve replacement (SAVR) in patients across the entire surgical risk spectrum. TAVR is now recommended as a possible therapeutic option for patients aged >65 years in the United States and 75 years in Europe with severe aortic stenosis (AS) and suitable clinical and anatomical characteristics.^{1,2} Recent data from large-scale randomized clinical trials even suggest extending its use to lower-risk patients.^{3,4} As a result, TAVR has become the predominant form of aortic valve replacement in Europe, as of 2018, and the United States.^{5,6}

Due to the rapid growth of the TAVR population, which includes younger patients with longer life expectancy, the durability of transcatheter heart valves

(THVs) has become a key issue in terms of cardiovascular death and morbidity in years to come. All bioprosthetic valves are susceptible to degeneration and, in common with surgical valves, THVs have a limited life span. Although THVs are similar in structure to surgical valves, it has been suggested that the lower risk of prosthesis–patient mismatch after TAVR compared with SAVR may result in longer durability.^{7–10} Conversely, others note several features of THVs that may limit their life spans, including interactions with the native valve and crimping of the THVs.

Several mechanisms can lead to bioprosthetic valve dysfunction (BVD) and eventually failure (BVF), including structural valve deterioration (SVD), thrombosis, endocarditis, and nonstructural valve deterioration (patient–prosthesis mismatch and paravalvular leak).¹¹

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Nonstandard Abbreviations and Acronyms

AS	aortic stenosis
ATLANTIS	Antithrombotic Strategy After Trans-Aortic Valve Implantation for Aortic Stenosis
BVD	bioprosthetic valve dysfunction
BVF	bioprosthetic valve failure
CHOICE	Randomized Comparison of Transcatheter Heart Valves in High-Risk Patients With Severe Aortic Stenosis: Medtronic CoreValve Versus Edwards SAPIEN XT
DOAC	direct oral anticoagulant
ENVISAGE-TAVI	Edoxaban Versus Standard of Care and Their Effects on Clinical Outcomes in Patients Having Undergone Transcatheter Aortic Valve Implantation
EOA	effective orifice area
GALILEO	Global Study Comparing a Rivaroxaban-Based Antithrombotic Strategy to an Antiplatelet-Based Strategy After Transcatheter Aortic Valve Replacement to Optimize Clinical Outcomes
HALT	hypoattenuated leaflet thickening
HVD	hemodynamic valve deterioration
NOTION	Nordic Aortic Valve Intervention
OAC	oral anticoagulant
PARTNER	Placement of Aortic Transcatheter Valves
PPM	patient–prosthesis mismatch
redo-TAVR	transcatheter valve-in-valve
SAPIEN 3	Valve Performance of the SAPIEN 3 Ultra RESILIA Valve: A Prospective Registry With Central Echocardiography Analysis
SAVR	surgical aortic valve replacement
SLT	subclinical leaflet thrombosis
SURTAIV	Surgical or Transcatheter Aortic-Valve Replacement in Intermediate-Risk Patients

SVD	structural valve deterioration
TAVR	transcatheter aortic valve replacement
THV	transcatheter heart valve
TTE	transthoracic echocardiography
VARC	Valve Academic Research Consortium
VIVID	Valve-in-Valve International Data
VKA	vitamin K antagonist

Endocarditis, thrombosis, and nonstructural valve deterioration are 3 factors related to pathological issues and are due to intrinsic structural changes to the bioprosthesis. They can be potentially treated with antimicrobial therapy or antithrombotic treatment, and specifically prevented using proper sizing and procedural techniques. By contrast, SVD remains a significant concern, with an incidence of up to 12.3% 5 years after TAVR. It represents the ultimate consequence of fibrotic remodeling and calcification of the bioprosthetic tissue. Unlike SAVR, the native aortic valve may interact with the bioprosthesis after TAVR, deforming the stent frame and driving thrombotic and inflammatory processes.¹² The higher incidence of hypoattenuated leaflet thickening (HALT) after TAVR compared with SAVR, and the association between HALT and SVD, illustrates this mechanism. Additional immunological and cellular pathways are also involved in the development of SVD. Understanding these pathomechanisms and predictive factors is crucial for future valve design and the development of potential adjunctive therapies for patients.

In this review, we seek to present an overview of the current data on the definitions, epidemiology, pathophysiology, clinical consequences, prevention, and management of SVD in THVs.

DEFINITIONS OF SVD

A comparison of the durability of different types of bioprosthesis requires a standardized terminology for the definition of a degenerated bioprosthetic aortic valve. The definition of SVD has evolved over the years as we have gained a better understanding of the various mechanisms involved.

Evolution of the Definition of SVD Over Time

Numerous observational studies have contributed with data regarding the durability of bioprostheses,

including several investigations on surgical replacements adopting freedom from reoperation as the principal end point for durability assessment. However, reinterventions may be undertaken for indications other than SVD, and some interventions might be non-performed in a significant proportion of patients due to their age, comorbidities, or frailty, and a considerable proportion of patients could face nonindicated procedures according to most recent echocardiographic criteria.^{11,13} Other studies on surgical bioprosthesis used definition based on reoperation, autopsy, or clinical investigation.¹⁴ These early definitions are limited, as they do not clearly describe the morphological and hemodynamic changes that occur during SVD.

Partly addressing this issue, the American Society of Echocardiography in 2009 released guidelines for the echocardiographic evaluation of bioprosthetic aortic valves.¹⁵ Following similar lines, the second Valve Academic Research Consortium (VARC-2) consensus from 2012 defined SVD as valve-related stenosis (with a mean aortic gradient ≥ 20 mmHg; effective orifice area (EOA) < 0.9 or 1.1 cm² according to body surface area; and a Doppler velocity index < 0.35 m/s), prosthesis–patient mismatch, or moderate to severe prosthetic valve regurgitation.¹⁶

Contemporary Definitions

The 2017 European Association of Percutaneous Cardiovascular Interventions consensus further refined SVD as intrinsic permanent changes of the bioprostheses (such as leaflet fibrosis, calcification, tear, or flail) leading to morphological and hemodynamic dysfunction. This definition clearly distinguished SVD from non-SVD, which includes changes not intrinsic to the bioprosthesis itself (intra- or paraprosthetic regurgitation and patient–prosthesis mismatch)¹³ (Figure 1). Furthermore, the VIVID (Valve-in-Valve International Data) study in 2018 defined SVD as a continuum between morphological and hemodynamic changes.¹⁷ Finally, the recent VARC-3 consensus integrated multimodality imaging to provide additional criteria to distinct SVD, non-SVD, thrombosis, and endocarditis.¹¹ According to this definition (Figure 1), SVD starts with morphologic changes (wear and tear, leaflet disruption, flail leaflet, fibrotic leaflet, calcification, or strut fracture), and could lead to moderate or severe hemodynamic valve deterioration (HVD). The VARC-3 definition is likely the most robust definition of SVD and this is thus the one that should be applied in future studies.¹⁸

Future Directions for Refining Definitions

While these definitions are mainly based on the hemodynamic assessment of bioprosthesis by echocardiography, previous studies demonstrated the discordance between post-TAVR echocardiographic and invasive

transvalvular gradients,^{19,20} and the nonlinear impact of echocardiographic gradient and clinical outcomes.²¹ According to these data, invasive hemodynamic assessment of bioprosthesis might be included in the definitions of SVD in the future in specific populations, especially in patients with inconclusive echocardiography or those with high risk of discordance between echocardiographic and invasive measures.²²

In summary, SVD is an evolving concept whose definition has changed in parallel with the evolution of aortic valve replacement techniques and imaging modalities. In its more recent definitions, SVD includes intrinsic morphologic changes leading to progressive hemodynamic abnormalities with clinical impact.

CURRENT DATA ON TRANSCATHETER HEART VALVE DURABILITY

When assessing THVs durability, it is important to keep data on the durability of surgical bioprostheses in mind. However, direct life span comparisons between these 2 types of bioprosthesis are limited by confounding factors related to comorbidities, such as age or chronic kidney disease, that are possibly associated with SVD. Various surgical bioprostheses have been developed and improved over several decades to enhance their durability. Differences in their composition may impart distinct characteristics to different bioprostheses, such as variations in the EOA, transprosthetic gradient, left ventricular remodeling, and overall durability.²³ Recent investigations have underscored substantial disparities among distinct categories of surgical bioprostheses highlighting a projected durability of at least 10 years for those endowed with a superior profile.²⁴ A previous study assessing the long-term outcomes of a real-world cohort of patients undergoing SAVR showed that the incidence of severe SVD was 6.6% at 10 years, while 30.1% of the population had moderate SVD.²⁵

Bench-Testing and Modeling Studies

Important insights have also been provided by bench-testing studies. In a previous research, accelerated wear testing equivalent to 1 billion cycles, which corresponds to ≈ 25 years of use, was performed to compare the durability of Sapien 3 valves (Edwards Lifesciences, Irvine, CA) with Magna Ease surgical valves (Edwards Lifesciences).²⁶ Similar durability results were observed for both varieties. In addition, a recent modeling study found that a THV would need on average to be only 30% as durable as a typical surgical bioprosthesis to achieve similar life expectancies with SAVR.²⁷ However, uncertainties persist for younger and lower-risk patients due to the potential

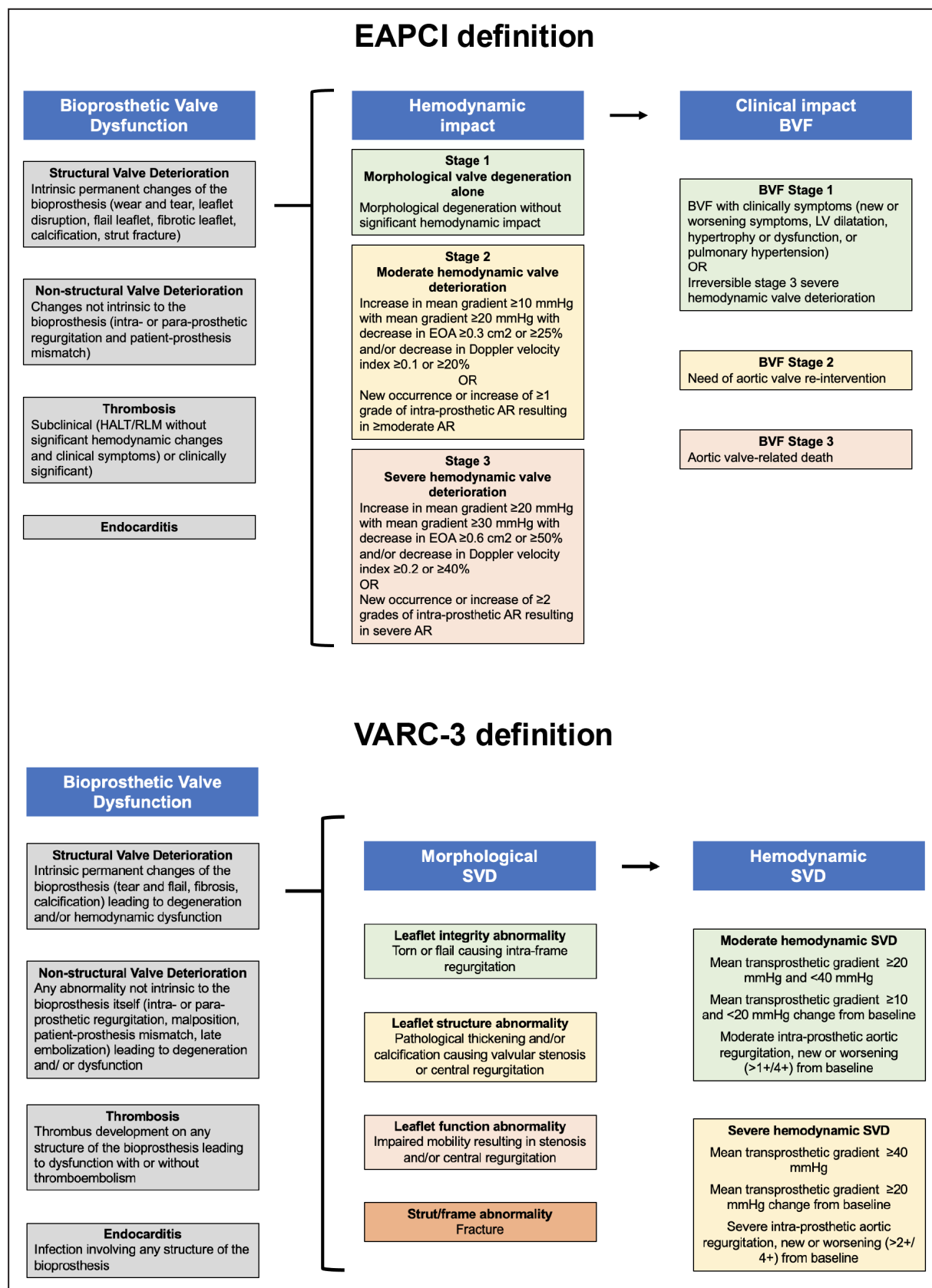


Figure 1. Summary of the main definition of structural valve deterioration.

AR indicates aortic regurgitation; BVF, bioprosthetic valve failure; EOA, effective orifice area; HALT, hypoattenuated leaflet thickening; LV, left ventricular; RLM, reduced leaflet motion; and SVD, structural valve deterioration.

implications of THV durability on their long-term outcomes. It is also essential to emphasize that although bench-testing and modeling studies provide valuable insights, they cannot achieve a full replication of the complexities of rheological and biological interactions and the dynamic environments within the human body. Therefore, both clinical studies and real-world data are required to complement and validate these findings.

Clinical Data

Long-term data concerning THV durability remain limited in comparison with surgical valves. This scarcity can be attributed to the relatively recent integration of TAVR into clinical practice, marked by its inaugural implantation in human subjects in 2002, followed by the initial regulatory approvals in 2011. Additionally, the limited life expectancy of the first patients undergoing TAVR further limits the availability of long-term data on THV durability. As a result, most data focus on the first 5 years following the procedure. It may be useful to know the incidence of non-SVD in comparison with SVD. Non-SVD is mainly represented by paravalvular leak, with an incidence of 7% to 40%,²⁸ which has decreased with recent THV designs, and patient–prosthesis mismatch, with an incidence of 0.1% to 5.7% in its severe form, depending on the measurement method.⁸ While paravalvular leak was negatively correlated with post-TAVR survival, data on the impact of patient–prosthesis mismatch on death are conflicting.

The principal research findings on SVD to date are summarized in the [Table 1](#). Studies with durability data at 5 years mainly included high-risk patients and demonstrated a cumulative incidence of BVD from 0.0% to 12.3%, with the majority of them <3.5%.^{29–37} Two post hoc analyses of large populations derived from randomized controlled trials deserve specific attention. First, the data from the PARTNER (Placement of Aortic Transcatheter Valves)-2A trial and the SAPIEN 3 (Valve Performance of the SAPIEN 3 Ultra RESILIA Valve: A Prospective Registry With Central Echocardiography Analysis) registry showed that the second-generation SAPIEN XT balloon-expandable valve has a higher 5-year rate of SVD according to VARC-3 definition ($1.61 \pm 0.24\%$) in comparison with SAVR ($0.63 \pm 0.16\%$, $P \leq 0.01$), while the third-generation SAPIEN 3 valve demonstrated a rate of SVD similar to SAVR in intermediate risk patients, both after propensity matching ($0.68 \pm 0.18\%$ versus $0.60 \pm 0.17\%$, respectively; $P=0.71$) and inverse probability treatment weighting adjustment ($0.78 \pm 0.17\%$ versus $0.64 \pm 0.18\%$, respectively; $P=0.37$).³⁵ Consistently, the recent data on the 5-year echocardiographic outcomes of the PARTNER-3 trial showed similar rate of SVD-related bioprosthetic valve failure in low-risk patients treated by TAVR and SAVR

(1.6% versus 2.4%, respectively; $P=0.42$).³⁷ Second, the recent analysis from the CoreValve US High Risk Pivotal and SURTAVI (Surgical or Transcatheter Aortic-Valve Replacement in Intermediate-Risk Patients) randomized clinical trials demonstrated a lower rate of SVD in intermediate- or high-risk patients undergoing TAVR with a self-expandable valve in comparison with SAVR (2.20% versus 4.38%, respectively; $P=0.004$) at 5 years.³⁶

Data extending beyond a 5-year interval post-TAVR remain limited and reported a rate of BVD from 0.0% to 12.7% at 10 years.^{38–48} It should be acknowledged that these different registries may have been impacted by mortality bias, which precludes the capture of all cases of valvular degeneration. To date, the NOTION (Nordic Aortic Valve Intervention) trial remains the only randomized clinical trial providing long-term comparative data on bioprosthesis durability in low-risk patients treated by TAVR or SAVR.⁴⁵ At 8 years, the risk of SVD was lower after TAVR than after SAVR (13.9% versus 28.3%, respectively; $P=0.0017$), while the risk of bioprosthetic valve failure was similar (8.7% versus 10.5%, respectively; $P=0.61$). The 10-year results of the NOTION trial have been recently published⁴⁹: The risk of severe SVD was lower after TAVR than after SAVR after 10 years (1.5% versus 10.0%, respectively; $P=0.02$). However, the interpretation of these data is limited by the use of first-generation self-expandable valves and the relatively low number of patients alive at 10 years after the procedure.

Altogether, these findings are reassuring regarding the long-term durability of THV compared with surgical bioprostheses. However, further long-term data and continuous improvement in THV technology remain essential to improve their durability, and understanding the underlying pathomechanisms involved in SVD is a critical aspect of this process.

PATHO-MECHANISMS OF STRUCTURAL VALVE DETERIORATION

Bioprosthetic heart valves mainly derived from bovine or porcine pericardium or, less frequently, from porcine heart valves. They are usually cross-linked with glutaraldehyde to ensure tissue stability, reduce antigenicity, and maintain tissue sterility. They undergo progressive degeneration over time, as a possible consequence of regenerative capability impairment, collagen degradation, or calcium precipitation by residual lipids. Active calcification is viewed as the final common pathway of SVD, leading to narrowing of the heart valve or tearing of the leaflets. Previous histological analyses emphasize that SVD is a dynamic process in which thrombosis, endothelialization, and inflammation modulate further valve remodeling, leading to fibrosis and calcification.^{50–53} These data highlight that thrombosis may

Table 1. Data on THV Durability

Study	Type of study	Number of patients	Type of patients (mean age, y)	Type of valve	SVD definition	Cumulative incidence of SVD-related BVF
Data at 5 y						
Toggweiler et al ²⁹ 2013	Monocentric registry	88	High risk (83.0)	Balloon-expandable	VARC-1	3.4%
Mack et al ³⁰ 2015	RCT	348	High-risk (84.1)	Balloon-expandable	Need of surgical explant	0.0%
Barbanti et al ³¹ 2015	Multicentric registry	353	High-risk (81.5)	Self-expandable	VARC-1	1.4%
Didier et al ³² 2018	Multicentric registry	4210	High risk (83.0)	Balloon- or self-expandable	EAPCI	2.5%
Gleason et al ³³ 2018	RCT	391	High risk (83.2)	Self-expandable	EAPCI	0.8%
Orvin et al ³⁴ 2019	Multicentric registry	450	High risk (82.2)	Balloon- or self-expandable	EAPCI	12.3%
Pibarot et al ³⁵ 2020	RCT	1665	Inter. risk (81.8)	Balloon-expandable	VARC-3	1.6% for Sapien XT 0.7% for Sapien 3
O'Hair et al ³⁶ 2023	RCT	1128	Intermediate and high risk (82.1)	Balloon-expandable	VARC-3	2.2%
Mack et al ³⁷ 2023	RCT	495	Low risk (73.3)	Balloon-expandable	VARC-3	1.6%
Data beyond 5 y						
Blackman et al ³⁸ 2019	Multicentric registry	241	High risk (79.3)	Balloon- or self-expandable	EAPCI	0.4% At 5.8 y
Deutsch et al ³⁹ 2018	Monocentric registry	300	High risk (81.4)	Balloon- or self-expandable	EAPCI	14.9% At 7 y
Durand et al ⁴⁰ 2019	Multicentric registry	1403	High risk (82.6)	Balloon- or self-expandable	EAPCI	4.2% At 7 y
Eltchaninoff et al ⁴¹ 2018	Monocentric registry	378	High risk (83.3)	Balloon- or self-expandable	EAPCI	3.2% At 8 y
Holy et al ⁴² 2018	Monocentric registry	152	High risk (81.0)	Self-expandable	EAPCI	0.0% At 8 y
Barbanti et al ⁴³ 2018	Monocentric registry	288	High risk (80.7)	Balloon- or self-expandable	EAPCI	2.4% At 8 y
Ferreira-Neto et al ⁴⁷ 2020	Monocentric registry	295	High risk (80.0)	Balloon-expandable	Specific [*]	9.3% At 8 y
Testa et al ⁴⁴ 2020	Monocentric registry	990	High risk (82.0)	Self-expandable	EAPCI	1.6% At 8 y
Jorgensen et al ⁴⁵ 2021	RCT	145	Low risk (79.1)	Self-expandable	EAPCI	2.2% At 8 y
Alaour et al ⁴⁸ 2025	Monocentric registry	2403	High risk (81.8)	Balloon- or self-expandable	VARC-3	12.7% At 10 y
Sathananthan et al ⁴⁶ 2021	Monocentric registry	235	High risk (82.4)	Balloon- or self-expandable	EAPCI	6.5% At 10 y

BVF indicates bioprosthetic valve failure; EAPCI, European Association of Percutaneous Cardiovascular Interventions; EOA, effective orifice area; RCT, randomized clinical trial; SVD, structural valve deterioration; and VARC, Valve Academic Research Consortium.

*The definition of SVD used in this study was specific. Subclinical SVD was defined as in the presence of an absolute increase in mean gradient >10 mm Hg with a decrease in EOA >0.3 cm² (and/or decrease in Doppler velocity index >0.08), and/or new onset of at least mild intraprosthetic aortic regurgitation or increase by at least 1 grade of preexistent intraprosthetic valve regurgitation, with the resulting regurgitation grade inferior or equal to moderate; or change in morphology (thickening, calcification, flail, pannus) and/or mobility (reduced, avulsed) of transcatheter heart valve leaflets. Clinically relevant SVD was defined as an increase in mean gradient >20 mmHg with a concomitant decrease in the EOA >0.6 cm² (and/or decrease in Doppler velocity index >0.15), generating a severe aortic stenosis according to current guidelines; and/or new occurrence or increase of at least 1 grade of intraprosthetic regurgitation leading to moderate-to-severe aortic regurgitation.

constitute an active nidus of biological effectors paving the way to SVD (Figure 2).

Thrombotic Pathway

The thrombotic burden after TAVR is of particular interest. Defined as the occurrence on multidetector computed tomography of HALT and reduced leaflet motion, subclinical leaflet thrombosis (SLT) may be detected in 6% to 15% of patients after TAVR and up to 30.9% after 1 year.^{54,55} While the first studies concerning its clinical and prognostic impact were reassuring, recent works indicate that HALT is associated with higher risk of 1-year all-cause death and cardiac death, heart failure hospitalization, cerebrovascular events, and SVD.^{12,56} The clinical data from Hein et al⁵⁶ demonstrating a significant association between HALT and SVD are coherent with the histopathological observations from Sellers et al⁵⁰ in which THV fibrosis and calcification were observed only in case of thrombosis development. Based on echocardiographic and histologic data,⁵⁷ some authors proposed that SVD is the result of a continuum that starts with SLT through a mechanism of thrombus-mediated bioprosthetic valve calcification.

Furthermore, a recent study underscored the connection between the deformation of THV and the occurrence of HALT.⁵⁸ In this work, the prosthesis deformation index (integrating diameters at inlet, outlet, and waist), asymmetric leaflet expansion degree, and

neosinus volume were highly correlated with HALT occurrence in case of balloon-expandable valve implantation. Among self-expandable valve, eccentricity at leaflet inflow and prosthesis waist, asymmetric leaflet expansion degree and neosinus volume, were associated with HALT. In another study, THV eccentricity assessed as a binary variable (threshold of 10% of eccentricity defined as $1 - \text{minimum THV external diameter}/\text{maximum THV external diameter}$) was not associated with SVD occurrence.⁴⁷ THV eccentricity thus appears as a complex variable with multiple combinations of asymmetrical expansion, which could occur at different levels of the THV (inflow, outflow, waist, or skirt). To note, underexpansion of the functional portion of THV was common after valve-in-valve TAVR in a recent work, occurring more frequently with deep implantation and in a polymer surgical stent frame, and was associated with impaired postprocedural hemodynamic.⁵⁹ Further studies based on computed tomography analyses are required to confirm the link between THV deformation and thrombosis. Optimizing the hemodynamic performance at the time of THV implantation by resolving any obvious eccentricity or deformation appears crucial.

The impact of THV geometry on thrombosis occurrence suggests the role of local interaction between the native aortic valve and the bioprosthesis. Previous works have emphasized the interaction between AS pathomechanisms and the hemostatic system, as well as the acquisition during the natural history of AS of a

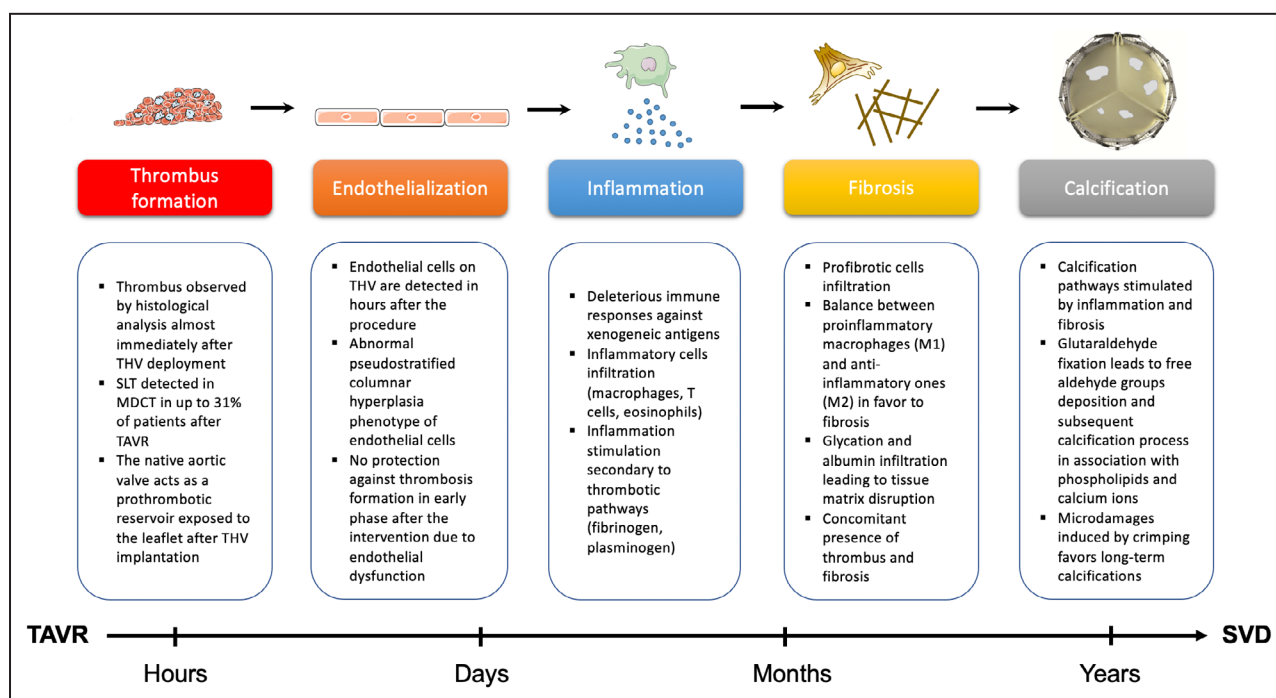


Figure 2. Pathomechanisms involved in structural valve deterioration after transcatheter aortic valve replacement.

MDCT indicates multidetector computed tomography; SLT, subclinical leaflet thrombosis; SVD, structural valve deterioration; TAVR, transcatheter aortic valve replacement; and THV, transcatheter heart valve.

prothrombotic phenotype of the native aortic valve.¹² The interaction between the diseased native aortic valve and the bioprostheses may explain the increased incidence of leaflet thrombosis at 1 month after TAVR compared with SAVR^{55,60} in which the native valve is removed. During THV deployment, the native valve may be injured and expose its prothrombotic content to the blood flow and the bioprosthesis, promoting thrombosis development.¹² Our team has recently demonstrated that a human calcified stenotic aortic valve constitutes an important reservoir of procoagulant extracellular vesicles, especially in the most diseased parts of the valve.⁶¹ Confined within the native valve, stenotic aortic valve extracellular vesicles acted as a potent biological agonist and transformed vascular endothelial cells into a prothrombotic, proadhesive and proinflammatory surface that recruited inflammatory cells and promoted thrombogenicity. Consistent with this paradigm, intra-annular valves together with oversizing may favor noxious interactions with trapped native valves and subsequent thrombus development.⁶²

Immune Pathway

The deleterious effect of immune responses against xenogeneic antigens could also contribute to SVD. The fixation of glutaraldehyde during the manufacture of bioprostheses leads to the deposition of free aldehyde groups, which results in a passive calcification process in association with phospholipids and calcium ions in the circulation, but also in a stimulation of humoral and cellular responses.⁶³ The pretreatment of commercial bioprostheses to reduce immunogenicity against animal-derived proteins does not guarantee total biocompatibility: some epitopes, including the xenoantigens galactose- α 1,3-galactose, and N-glycolylneuraminic acid, have been previously detected in bioprostheses and demonstrated their impact in bioprosthesis calcification.⁶⁴ Several sets of data have also demonstrated the impact of infiltrated M2 macrophages,⁶⁵ glycation and albumin infiltration,⁶⁶ and excessive accumulations of erythrocytes, circulating proteins, and CD68-positive macrophages and their activators, such as fibrinogen and plasminogen.⁶⁷

Role of the Circulating Endothelial Progenitor Cells

Circulating endothelial progenitor cells (EPCs) may also play a central role in the pathophysiology of SVD. Previous studies have demonstrated a higher rate of EPCs expressing the osteoblastic cell surface marker osteocalcin and a reduced number of reparative EPCs in patients with endothelial dysfunction related to both coronary atherosclerosis and AS.⁶⁸ Given the common mechanisms involved in AS and SVD, a recent study

investigated the contribution of EPCs expressing osteocalcin in SVD.⁶⁹ The authors found that patients who developed SVD after bioprosthetic aortic valve replacement had lower levels of reparative CD34⁺ CD133⁺ CPCs, which are essential for re-endothelialization, neovascularization, and repair of endothelial function. In addition, they observed higher levels of osteocalcin-positive EPCs, which may contribute to bioprosthesis calcification.

Leaflet Mechanical Stress

In addition to these biological mechanisms, THVs undergo different mechanical stress in hydrodynamic conditions, including shear stress, bending deformation, and leaflet tension.⁷⁰ The cyclic hemodynamic load applied on THVs may lead to leaflets delamination and calcification.⁷¹ The final THV geometry after implantation is correlated with the mechanical durability and the calcification rate.⁷² Balloon postdilation may also cause damage to the THV leaflets or to the stent frame.⁷³ According to an ex vivo bench study, excessive THV overexpansion has been associated with impaired hydrodynamic function, acute leaflet failure, and reduced durability.⁷⁴ Similarly, underfilling has been associated with higher transvalvular gradients, especially in smaller THV sizes, which may impair long-term valve durability.⁷⁵ Finally, TAVR valves are subject to significant compression during crimping, when the outer diameter of the bioprosthesis is reduced down to 14 to 18 Fr to ensure the passage through the introducer sheath.⁷⁶ Crimping exerts immediate, significant, and definitive structural damages to the bioprosthesis,⁷⁷ favoring long-term calcification.⁷⁸

CLINICAL CONSEQUENCES OF SVD

Reports on the clinical outcomes of SVD remain conflicting. According to the VARC-3 consensus, the clinical consequences of SVD should be categorized using BVF staging system¹¹: Stage 1 includes any BVD meeting clinically significant criteria, stage 2 corresponds to aortic valve reintervention, and stage 3 represents valve-related death. However, stage 3 BVF is rarely reported in studies, as distinguishing valve-related death from other cardiovascular causes is challenging. More data are available on stage 2 BVF (aortic valve reintervention). Overall, the reintervention rate among patients undergoing TAVR is relatively low, not exceeding 10% in multiple studies.^{35,49,79} Additionally, a recent study found no significant association between BVD and increased all-cause or cardiovascular death.⁴⁸ However, it is essential to acknowledge a potential survival bias in this older and frail population, which may influence reported outcomes.

POPULATIONS AT RISK

The predictive factors for SVD following either SAVR or TAVR can be classified into characteristics related to the patient and those related to the procedure itself (Table 2).

Patient-Related Factors

Data on the association of age and sex with SVD are conflicting. Age at the time of intervention was a key factor in bioprosthesis durability after SAVR.⁸⁰ Bias may potentially exist due to the differential rate of redo surgery according to age, which favors the detection of SVD in younger individuals. A previous report noted an increased susceptibility of THVs to accelerated SVD in young patients, characterized by enhanced calcification, altered collagen microstructure with loss of alignment and increased crimp periods, and increased crosslinking.⁸¹ Regarding sex, male patients were associated with a higher rate of SVD in some studies,⁸² whereas others found that female sex was a risk factor for SVD after SAVR.⁸³

Salaun et al identified 2 types of HVD risk factors after SAVR according to length of time since intervention.⁸³ Factors independently associated with HVD occurring within the first 5 years after AVR included diabetes, smoking, renal insufficiency, baseline postoperative mean gradient ≥ 15 mmHg, at least mild transprosthetic regurgitation, and the use of stented rather than stentless bioprostheses. Factors associated with

HVD occurring after 5 years were female sex, warfarin use, stented bioprostheses, and severe patient–prosthesis mismatch. Body mass index was also positively correlated with the risk of HVD.⁸⁴ After TAVR, the lack of anticoagulation was associated with a significant increase of transvalvular gradients and a higher risk of HVD during follow-up,⁸⁵ suggesting a pathophysiological link between thrombosis and BVD. Overtchouk et al also suggested a beneficial effect of anticoagulation, with a lower rate of BVD after 3 years in patients undergoing anticoagulation therapy (adjusted odds ratio, 0.54 [95% CI, 0.35–0.82]; $P=0.005$).⁸⁶ Finally, a large multicenter registry including 1521 patients undergoing TAVR identified the absence of anticoagulation at hospital discharge as an independent predictor of HVD.⁸⁴

Others authors have stressed the importance of a dysmetabolic profile with elevated lipoprotein-associated phospholipase A2 activity, a high level of proprotein convertase subtilisin/kexin 9, and insulin resistance associated. These features were associated with a higher risk of HVD of surgically implanted bioprostheses.⁸⁷

Procedural Factors

Data are also available on the risk of SVD according to procedural characteristics. Little valve size⁴⁰ and the valve-in-valve procedure were associated with higher risk of BVD.^{40,84} Patient–prosthesis mismatch (PPM) is a cause of non-SVD and may contribute to the development of SVD. PPM occurs when the EOA of a

Table 2. Predictors of Structural Valve Deterioration After Bioprosthetic Aortic Valve Replacement

Predictors		Adjusted HR (95% CI; <i>P</i> value)
Patient-related characteristics	Age	Inconclusive*
	Sex	Inconclusive†
	Body mass index	1.08 (1.03–1.13; 0.001) ⁸⁴
	Diabetes	1.33 (1.06–1.66; 0.01) ⁸³
	Smoking	2.58 (1.85–3.60; <0.001) ⁸⁸
	Chronic kidney disease	1.10 (1.03–1.16; 0.047) ⁸²
	Dysmetabolic profile	1.15 (1.04–1.26; 0.004)
	Lp-PLA2 activity (nmol/min per mL)	4.36 (1.25–14.02; 0.01)
	PCSK9 >305 ng/mL	3.30 (1.19–9.23; 0.02) ^{‡,87}
Procedure-related characteristics	Insulin resistance	
	Lack of anticoagulation	3.35 (1.57–7.13; 0.002) ⁸⁴
	Severe patient-prosthesis mismatch	1.85 (1.12–2.87; 0.02) after SAVR ⁸⁸ 1.70 (0.9–2.9; 0.08) after TAVR ¹⁰
	Baseline postoperative mean gradient ≥ 15 mmHg	1.30 (1.05–1.62; 0.02) ⁸³
	THV size ≤ 23 mm	2.07 (1.14–3.76; 0.016) ⁸⁴
	Commissural misalignment	1.03 (1.01–1.04; 0.01) ^{§,95}

HR, hazard ratio; Lp-PLA2, lipoprotein-associated phospholipase A2; PCSK9, proprotein convertase subtilisin/kexin 9; SAVR, surgical aortic valve replacement; SVD, structural valve deterioration; TAVR, transcatheter aortic valve replacement; and THV, transcatheter heart valve.

*Bias may exist due to the differential rate of reintervention according to age, which favors the detection of SVD in younger individuals.

†Male sex was associated with a higher risk of SVD in 1 study (HR, 2.17 [95% CI, 1.53–3.12]; $P<0.0001$),⁸² whereas others found that female sex was a risk factor for SVD after SAVR.⁸⁰

‡Odds ratio values.

§Per increase of 10-degree misalignment.

normally functioning prosthesis is too small for the patient's body size. According to the VARC-3 consensus, PPM is moderate if EOA is between 0.65 and 0.85 cm²/m² (or 0.55 and 0.70 cm²/m² in patients with obesity) and severe if it is ≤ 0.65 cm²/m² (or ≤ 0.55 cm²/m² in patients with obesity).¹¹ The association between PPM and SVD has been previously documented following bioprosthetic SAVR.⁸⁸ Recently, PPM has been linked to a significantly increased risk of valve thrombosis and a trend toward an elevated risk of SVD after TAVR.¹⁰ In this study, the main predictors of PPM included a small annulus, valve-in-valve TAVR, and the use of balloon-expandable valves. Notably, these factors are also recognized as predictors of leaflet thrombosis.¹² The close interaction between prothrombotic material in the native aortic valve and the bioprosthesis is particularly relevant when balloon-expandable intra-annular valves are used or in patients with a small annulus.⁸⁹ These situations may increase the risk of thrombosis owing to altered flow dynamics and increased susceptibility to turbulent blood flow. Further studies are required to fully understand the connection between PPM and SVD. Identifying patients at risk by calculating the predicted EOA from the normal EOA reference value for the model and size of THV,⁹⁰ and optimizing procedural techniques with possible balloon predilation or postdilation^{91,92} are crucial steps to prevent PPM.

Moreover, commissural alignment is a recent concept and will probably become one of the remaining challenges, especially in terms of improving coronary access. Recent data indicate that significant commissural misalignment creates nonphysiological flow⁹³ leading to leaflet stress and blood stagnation. This condition favors increased thrombogenicity⁹⁴ and BVD, as demonstrated by the association between commissural misalignment and midterm transprosthetic gradient elevation following Sapien 3 valve implantation.⁹⁵

The impact of THV design on long-term durability remains a subject of debate. Landmark randomized controlled trials have reported similar hemodynamic performance between intra-annular and supra-annular designs compared with surgical bioprostheses.^{96–98} Notably, in the CHOICE (Randomized Comparison of Transcatheter Heart Valves in High-Risk Patients With Severe Aortic Stenosis: Medtronic CoreValve Versus Edwards SAPIEN XT) trial, the 5-year cumulative incidences of BVF were similar between intra-annular and supra-annular valves (4.3% versus 3.4%, respectively).⁹⁸ However, hemodynamic parameters and rates of HVD favored supra-annular devices (0.0% versus 6.6%, respectively; $P=0.018$). Observational data further support the hemodynamic advantages of supra-annular valves, with studies reporting higher indexed EOA and lower mean aortic gradients compared with an intra-annular one.⁹⁹ Another study found that intra-annular THVs were associated with increased mean

transaortic gradients, as well as higher rates of BVF and severe HVD.¹⁰⁰ The influence of THV design appears particularly significant in patients with a small annulus. In the SMART (Small Annuli Randomized to Evolut or SAPIEN) trial, a self-expanding supra-annular valve was superior regarding bioprosthetic valve dysfunction rates within the first year after TAVR.¹⁰¹ The long-term durability of different THV designs remains to be fully elucidated.

TREATMENT AND PREVENTION OF STRUCTURAL VALVE DETERIORATION IN TRANSCATHETER AORTIC VALVES: CURRENT GUIDELINES AND PERSPECTIVES

As with AS, no medical treatment has been validated to treat or halt the progression of SVD. Transcatheter valve-in-valve (redo-TAVR) and surgical explants have been proposed for patients with THV deterioration. So far, data on the efficacy and safety of these treatments in the context of THVs are limited. Since the number of patients with transcatheter SVD is anticipated to increase in the coming years, intense research is ongoing to develop strategies to prevent it.

Medical Approaches for Preventing Transcatheter Aortic Valve SVD

Given the relationship between SLT and SVD, previous research has investigated the effect of antithrombotic treatment on BVD incidence. Anticoagulation has been independently associated with a lower rate of BVD following TAVR in 2 observational studies.^{85,86} However, conflicting evidence exists, as another work suggested that oral anticoagulants (OACs) may accelerate THV calcification.⁴⁸ It seems crucial to distinguish between direct oral anticoagulants (DOACs) and vitamin K antagonists (VKAs). VKAs inhibit the activation of matrix Gla protein, a key endogenous inhibitor of calcification. This mechanism has been involved in the faster progression of AS severity in patients on VKAs.¹⁰² Randomized clinical trials investigating the liberal use of OACs after TAVR have failed to show any benefit from this strategy. The GALILEO (Global Study Comparing a Rivaroxaban-Based Antithrombotic Strategy to an Antiplatelet-Based Strategy After Transcatheter Aortic Valve Replacement to Optimize Clinical Outcomes) trial randomized 1,644 patients without OAC indication to low-dose rivaroxaban (10 mg/day) plus aspirin for 3 months followed by rivaroxaban alone versus aspirin and clopidogrel for 3 months followed by aspirin alone.¹⁰³ This trial was stopped prematurely because the rivaroxaban strategy was associated with a

higher risk of all-cause death, thromboembolic events, and bleeding events. Interestingly, a substudy of the GALILEO trial observed a lower SLT rate in the rivaroxaban group compared with the antiplatelet group.¹⁰⁴

The ENVISAGE-TAVI (Edoxaban Versus Standard of Care and Their Effects on Clinical Outcomes in Patients Having Undergone Transcatheter Aortic Valve Implantation) trial demonstrated the noninferiority of edoxaban versus VKAs in patients with atrial fibrillation undergoing TAVR regarding the composite of all-cause death, myocardial infarction, ischemic stroke, systemic thromboembolism, valve thrombosis, or major bleeding events.¹⁰⁵ More recently, the ATLANTIS (Antithrombotic Strategy After Trans-Aortic Valve Implantation for Aortic Stenosis) trial compared the use of apixaban with antiplatelets alone or VKAs according to the presence or not of an indication for OAC.¹⁰⁶ While apixaban was not superior to the standard of care in the clinical outcomes, this strategy reduced SLT in the majority of patients undergoing TAVR without established indication for anticoagulation.¹⁰⁷

Overall, it is important to note that no clear clinical benefit of anticoagulation for the prevention of SLT has been demonstrated so far. While anticoagulant strategies may help mitigate the risk of leaflet thrombosis and potentially reduce the incidence of BVD, these benefits must be carefully weighed against the increased risk of bleeding events, particularly in frail, older, and comorbid populations, where the bleeding risk may outweigh any theoretical advantages. Recent research by our group has demonstrated that the inhibition of secondary hemostasis by OAC over primary hemostasis disorder may hamper the hemostatic process extensively, resulting in an increase in the prevalence of bleeding complications.¹⁰⁸ Finding the optimal antithrombotic regimen after TAVR involves striking a balance between reducing the thrombotic burden and minimizing the bleeding risk.¹⁰⁹ On a pragmatic approach, point-of-care assays such as closure time with adenosine diphosphate monitoring may serve as a useful tool to tailor and adjust the risk of antithrombotic therapy after TAVR.¹¹⁰ The current guidelines on antithrombotic treatment following TAVR¹¹¹ are summarized in [Figure 3](#). For patients without a concurrent indication for OAC, a single antiplatelet strategy is recommended if no percutaneous coronary intervention has been recently performed, and routine use of OAC is not indicated. If percutaneous coronary intervention was performed within the past 3 months, dual antiplatelet therapy is suggested for a duration tailored to patient and interventional characteristics, but it should be as short as possible (1–6 months) due to the high bleeding risk.¹¹¹ For patients without another indication for OAC, further studies are required to explore alternative strategies, including the use of OAC at a lower dose or for a short period (as is frequently done after SAVR), followed by either no treatment or single

antiplatelet therapy. In patients with a concurrent indication for anticoagulation, OAC should be continued after TAVR. If recent percutaneous coronary intervention was performed, single antiplatelet therapy should be added for 1 to 6 months, depending on patient and interventional characteristics. The choice between DOACs and VKAs in this context remains controversial. While several observational studies concluded that DOACs are associated with lower mortality rates and fewer bleeding events than VKAs,^{112–115} the randomized controlled trial ENVISAGE-TAVI AF found a higher incidence of bleeding with edoxaban compared with VKA.¹⁰⁵ Interestingly, patients who received a low dose of edoxaban (30 mg once daily) demonstrated a similar incidence of major bleeding compared with those treated with VKA. This finding suggests that a lower dose of DOAC may help mitigate the late bleeding risk following TAVR.

The clinical management of SLT after TAVR ([Figure 3](#)) remains incompletely addressed by current guidelines. According to the European Society of Cardiology 2021 guidelines on valvular heart diseases,¹ anticoagulation should be considered in patients with HALT and reduced leaflet motion leading to elevated gradients, at least until resolution. Based on available data from randomized clinical trials,^{104,105,107} apixaban, rivaroxaban, or VKAs should be preferred. We recommend maintaining anticoagulation for 3 to 6 months, as this duration has been shown in the literature to resolve SLT in the majority of patients.^{54,60} If resolution of SLT is confirmed by follow-up cardiac computed tomography, the continuation of anticoagulation should be discussed, considering patient preferences and bleeding risk. If the cardiac computed tomography shows persistent SLT, adherence to treatment should be checked, and switching anticoagulation (eg, to another DOAC or VKA with an increased target international normalized ratio) should be considered.

Enhancing Durability through Bioprosthesis Development

In the pursuit of enhancing long-term durability, the optimal THV should inherently encompass fundamental attributes akin to those found in the native valve. These include the ability for self-reparation, adaptive remodeling, protection against infective agents, and the absence of thrombogenic tendencies. Improving the fixation process used to manufacture and store bioprostheses may also contribute to reducing the risk of SVD. Animal tissues are historically treated with glutaraldehyde to ensure they are immunologically inert, sterilized, and preserved. However, it has been demonstrated that the glutaraldehyde fixation plays a significant role in the calcification process, with residual aldehydes expressed on the tissue surface acting as

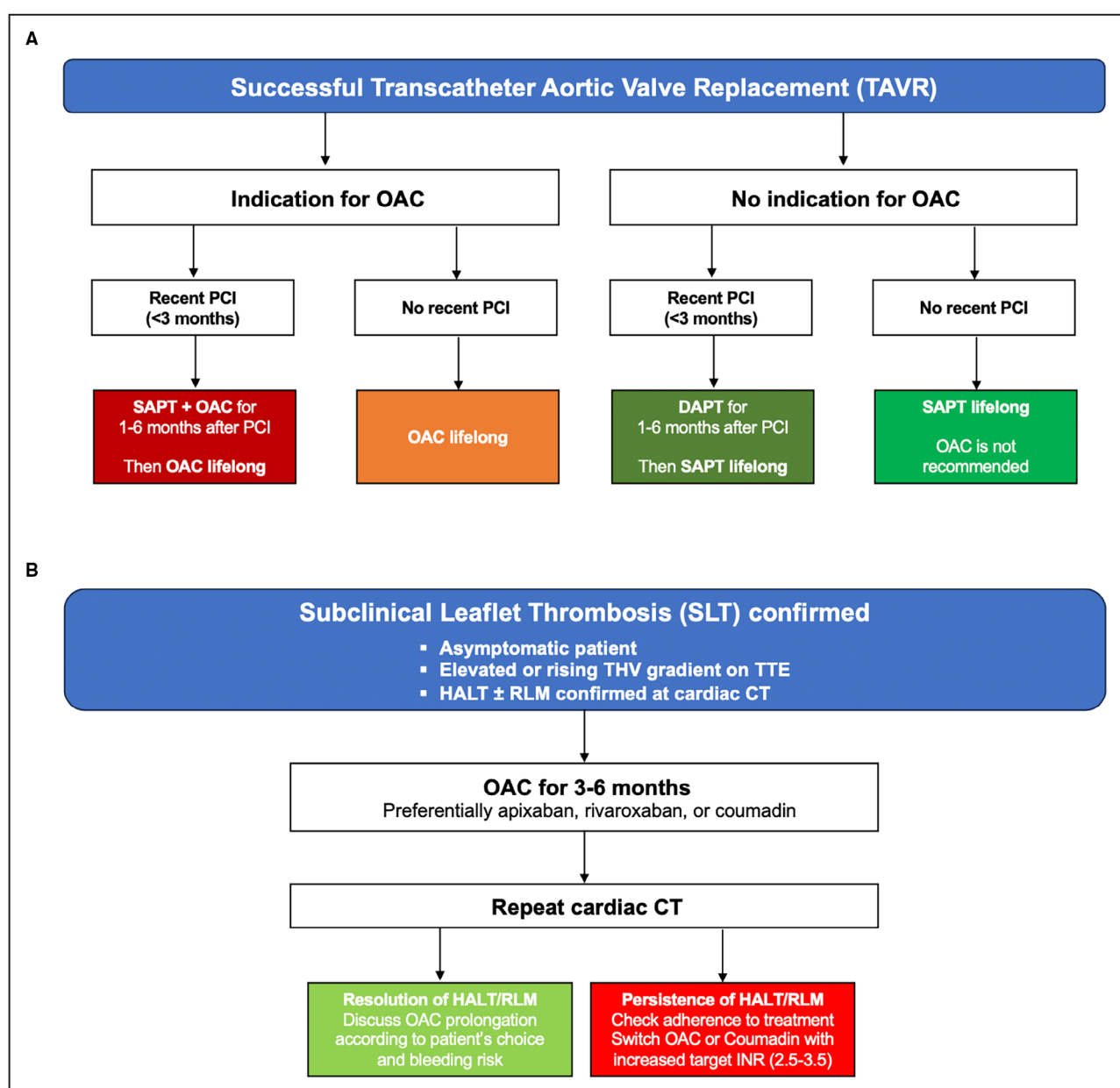


Figure 3. Current guidelines on antithrombotic treatment after transcatheter aortic valve replacement (A) and in patients with subclinical leaflet thrombosis (B).

CT indicates computed tomography; DAPT, dual antiplatelet therapy; HALT, hypoattenuated leaflet thickening; INR, international normalized ratio; OAC, oral anticoagulant; PCI, percutaneous coronary intervention; RLM, reduced leaflet motion; SAPT, single antiplatelet therapy; SLT, subclinical leaflet thrombosis; TAVR, transcatheter aortic valve replacement; THV, transcatheter heart valve; and TTE, transthoracic echocardiography.

focus points for calcification.¹¹⁶ In addition, the conventional glutaraldehyde crosslinking and anticalcification protocols of cardiac xenografts fail to totally avoid immune rejection of bioprostheses,⁶⁴ and the deleterious effects of tissue phospholipids, conformational changes in collagen, and free aldehyde groups have been observed.¹¹⁷

An alternative strategy using valves without galactose- α 1,3-galactose but adopting multiple anticalcification therapies including decellularization, immunologic modification

with α -galactosidase, space filler, organic solvent, and detoxification have demonstrated their preclinical safety and efficacy in a large animal model.¹¹⁸ Complementary strategies have been proposed to prevent calcification of the glutaraldehyde-fixed bioprosthesis such as calcitonin or sodium bisulfite treatments.^{119,120}

New technological refinements were recently provided by RESILIA tissue. This tissue is a bovine pericardial tissue incorporating a novel integrity preservation technology that includes stable capping of

free aldehydes, which prevents calcium binding, and glycerolization, which further prevents exposure to aldehydes and calcification. A randomized study in a chronic juvenile sheep model has demonstrated a 72% reduction of calcified tissue with RESILIA with respect to a contemporary bovine pericardial valve.¹²¹ Recently published, the 5-year outcomes of the COMMENCE Aortic trial that evaluated SAVR with RESILIA tissue in a contemporary SAVR population, have demonstrated low echocardiographic mean gradients over time, little transvalvular regurgitation, and no events of SVD.¹²² Accordingly, a nonrandomized comparison of RESILIA tissue versus contemporary aortic bioprostheses has suggested a reduction of SVD in RESILIA-treated valves (1% versus 4%).¹²³ The new version of the Sapien 3 Ultra valve powered by RESILIA tissue is available in numerous countries including the United States and is expected to result in a sizeable reduction of SVD.

Other alternative strategies have been advanced to optimize bioprosthesis manufacture. While fixed pericardial tissue is commonly used for available valves, decellularized xenograft tissues present the advantage of remodeling and growing and have the capacity to preserve tissue viability and function through recellularization.¹²⁴ Promising data for decellularization showed a 89% reduction in calcium content after 180 days in the mitral position of a sheep model.¹²⁵

Multimodality Imaging to Detect and Evaluate Transcatheter Aortic Valve SVD

If SVD occurs despite preventive measures, early diagnosis is crucial. The Heart Valve Collaboratory recently proposed a 4-step algorithm for detecting and evaluating BVD (Figure 4).

The first step involves assessing changes in the structure and hemodynamic function of the bioprosthesis by comparing baseline transthoracic echocardiography (TTE), which should be performed between 1 and 3 months after the procedure, with follow-up TTE, conducted annually or when new or worsening symptoms arise. BVD should be suspected if changes in leaflet mobility or thickening are observed, or if abnormal hemodynamic parameters are detected according to current definitions (Figure 1). Assessing the THV gradient may be particularly challenging after TAVR due to discrepancies between echocardiographic and invasive measurements.^{19,20} In most cases, the evolution of noninvasive gradient over time is sufficient to detect BVD. However, in specific cases, such as unexplained elevated gradients, invasive hemodynamic assessment may be considered.

The second step is to determine the pathogenesis of BVD among SVD, non-SVD, thrombosis and endocarditis. These conditions can coexist or occur sequentially with SVD, given their interrelation. A

multimodality imaging approach including TTE, transesophageal echography, computed tomography, and positron emission tomography is essential for identifying the different characteristics of SVD, such as leaflet calcification and leaflet thickening affecting motion. Computational modeling of blood flow using computational fluid dynamics simulations is gaining attention in the field of aortic valve disease and replacement.¹²⁶ Computational fluid dynamics simulations can predict hemodynamic parameters after aortic valve replacement, including paravalvular leakage,¹²⁷ wall-shear stress in the aorta based on valve design and size,¹²⁸ and flow patterns generated by different prosthesis types.¹²⁹ Additionally, artificial intelligence solutions, such as deep learning, may enhance the ability to compute relevant hemodynamic parameters, optimizing modeling-based treatment support.¹³⁰

Once SVD is diagnosed, the third and fourth steps involve staging the severity of its structural and hemodynamic valve deterioration using TTE, and assessing its clinical impact on the basis of current criteria (Figure 1). This staging process is critical for adapting treatment and making decisions regarding aortic valve reintervention indication and timing.

Managing Transcatheter Aortic Valve SVD

Therapeutic strategies for transcatheter SVD include surgical explant or a transcatheter approach with redo-TAVR. The choice between these 2 strategies depends on the underlying mechanisms of THV dysfunction, patient anatomy, clinical situation, and procedural risk. Two retrospective registry studies have highlighted the notable risk of 30-day and 1-year death with the surgical option.^{131,132} According to an analysis of the FRANCE 2 and FRANCE TAVI registries, reintervention is rare after TAVR (1.7% among 72 850 patients).¹³³ In this study, reintervention occurred mostly early after the procedure and by redo-TAVR.

While redo surgery may be technically challenging and carries a higher mortality risk, it is important to note the risk of selection bias in related research. The first data available on redo-TAVR are derived from a multicenter study analyzing the outcomes of 50 patients. Redo-TAVR was safe to treat postprocedural and late THV failure, and was associated with favorable clinical outcomes and valve performance.¹³⁴ In another study,¹³⁵ Landes et al demonstrated favorable results for redo-TAVR, with a survival rate of 94.6% at 30 days and 98.5% at 1 year for patients with late valve dysfunction. More recently, Tang et al have reported the results of the EXPLANTORREDO-TAVR (Explant or Redo-TAVR) registry. This study included 396 patients with either TAVR-explant (181 patients) or redo-TAVR (215 patients) for THV failure at a separate time from the initial TAVR.¹³⁶ Compared with redo-TAVR,

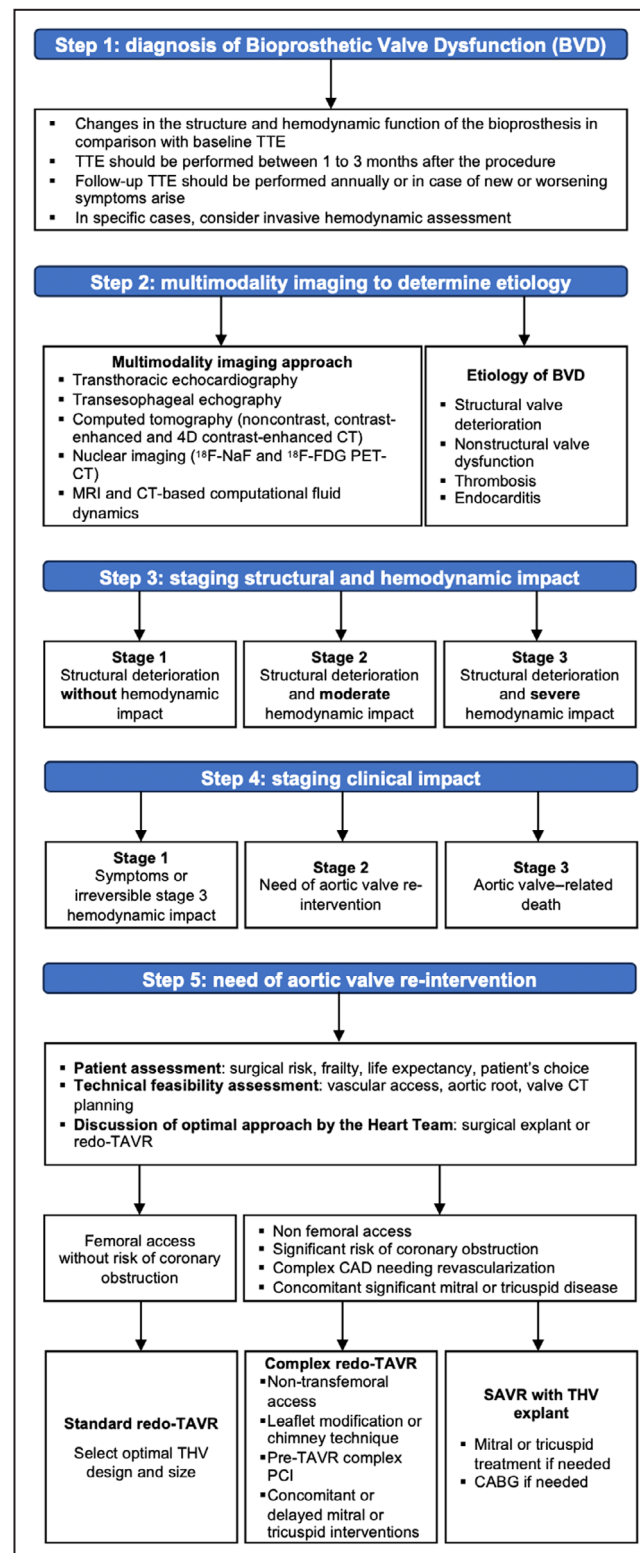


Figure 4. Diagnosis, exploration, and management of structural valve deterioration after TAVR.

BVD indicates bioprosthetic valve dysfunction; CABG, coronary artery bypass grafting; CT, computed tomography; MRI, magnetic resonance imaging; PCI, percutaneous coronary intervention; PET, positron emission tomography; TAVR, transcatheter aortic valve replacement; THV, transcatheter heart valve; and TTE, transthoracic echocardiography.

TAVR-explant had a higher 30-day (13.6% versus 3.4%; $P<0.001$) and 1-year mortality rate (32.4% versus 15.4%; $P=0.001$). In the landmark analysis, the mortality rate beyond 30 days was not significantly different between redo-TAVR and surgical explant.

Despite these reassuring data on redo-TAVR, we should not dismiss the risk of coronary obstruction, which remains the main challenge of such procedure. Bench testing has shown that most designs of THV and procedural strategies for redo-TAVR result in excellent bioprostheses performance afterwards,¹³⁷ while certain THV design and implantation positions seem to be more favorable. A redo-TAVR with a tall-frame self-expanding valve within another tall frame requires perfect cell alignment to ensure optimal coronary access.¹³⁸ Leaflet modification strategies to prevent coronary obstruction during valve-in-valve TAVR procedures have been developed and demonstrated their feasibility.^{139,140} Long-term data on their safety and efficacy are required to use these techniques in daily practice.

The selection of a therapeutic strategy among the available options, including redo-TAVR, SAVR with THV explant, or medical management, should be guided by a comprehensive patient assessment. This assessment should consider surgical risk, frailty, life expectancy, and patient preferences. Additionally, the technical feasibility of the procedure, such as vascular access, aortic root characteristics, and valve computed tomography planning should be evaluated. The presence of significant cardiac comorbidities, including complex coronary artery disease requiring revascularization or concomitant mitral or tricuspid disease, should also play a crucial role in decision making. Finally, the concept of “lifetime management” for severe AS is critically important, especially for younger patients who may require more than one aortic valve replacement over their lifetime. A significant risk of coronary obstruction does not permit redo-TAVR in all patients and should be anticipated before the initial procedure. This information is crucial for guiding patient decision making, potentially favoring SAVR first with the option of valve-in-valve TAVR in the future.

CONCLUSIONS

In the rapidly evolving landscape of TAVR, it is crucial to acknowledge the emerging and significant concern surrounding THV durability. As the candidate population for TAVR expands, encompassing younger patients with extended life expectancies, the spotlight intensifies on THV longevity. While the current data provide reassurance regarding midterm THV durability in comparison with surgical bioprostheses, we need more long-term data. It remains imperative that future

efforts be directed toward a better understanding of SVD pathophysiology and novel approaches to analyze phenotypic features of patients associated with SVD to improve and refine preventive and management strategies for patients with THV.

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